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ORIGINAL ARTICLE

Adenoviral GDNF gene transfer enhances neurofunctional recovery after recurrent laryngeal nerve injury

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To assess the possibility of gene therapy for recurrent laryngeal nerve (RLN) injury, we examined functional and histological recovery after glial cell line-derived neurotrophic factor (GDNF) gene transfer in a rat RLN crush model. Adenoviral vector encoding β-galactosidase gene (AxCA-LacZ) or human GDNF gene (AxCAhGDNF) was injected into the crush site of the RLN. Neurons in the nucleus ambiguus on the crushed side were labeled with X-gal or GDNF immnohistochemistry after AxCALacZ or AxCAhGDNF injection. Reverse transcription-polymerase chain reaction analysis revealed expression of human GDNF mRNA transcripts in brainstem tissue containing the nucleus ambiguus on the crushed side after AxCAhGDNF injection. Animals injected with AxCAhGDNF displayed significantly improved motor nerve conduction velocity of the RLN and recovery rate of vocal fold movement at 2 and 4 weeks after treatment as compared to controls. AxCAhGDNF-injected animals showed a significantly larger axonal diameter and improved remyelination in crushed RLN as compared to controls. Adenoviral GDNF gene transfer may thus promote laryngeal function recovery after RLN injury. Inoculation of adenoviral vector containing the GDNF gene at the site of damage soon after nerve injury may assist patients with laryngeal paralysis caused by nerve injury during head and neck surgery.
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Keywords: *adenovirus; cranial nerve injury; GDNF; nucleus ambiguus; recurrent laryngeal nerve; vocal fold palsy*

Introduction

The recurrent laryngeal nerve (RLN) comprises neurites of motor, sensory and autonomic neurons. One of the most significant laryngeal impairments caused by RLN injury is incomplete glottic closure, which induces dysphonia, dysphagia and aspiration pneumonia. These symptoms are associated with decreased quality of life. RLN injury can be caused by direct invasion of malignant tumor, such as thyroid or esophageal cancer, surgical ablation of the neck or mediastinum, thoracic aortic aneurism and intubation.

Medialization procedures (vocal fold injection, thyroplasty and arytenoid adduction) are commonly performed to manage unilateral vocal fold palsy, and can improve symptoms by enhancing glottic closure. However, these only achieve vocal fold medialization by static changes in vocal fold tissue or the laryngeal framework, and such deficits can never be neurologically restored.[1] Voice therapy is effective only for mild cases and reinnervation procedures have been attempted in very few cases.[2–4] Reinnervation procedures (neural anastomosis, nerve graft, etc.) for the RLN may improve voice quality through restoration or preservation of laryngeal muscle tone and have the potential to restore vocal fold mobility.[5,6] However, this procedure appears to have had little impact on the recovery of dynamic laryngeal function and is not widely accepted as a treatment option. The failure of reinnervation procedures is the result of multiple factors, including decreased motor fiber density, existing laryngeal muscle atrophy, central motoneuron loss in the nucleus ambiguus, and inappropriate or misdirected innervation by antagonistic motoneurons.[7,8]

The discovery of neurotrophic factors that support neuronal survival during development and neuronal function throughout adulthood has generated broad interest in the use of these factors in facilitating recovery after peripheral nerve injury. A multitude of neurotrophic factors have been shown to prevent motoneuron loss, enhance nerve sprouting, preserve motor endplate morphology, and promote reinnervation.[9–15] For example, glial cell line-derived neurotrophic factor (GDNF), brain-derived neurotrophic factor, neurotrophin (NT)-3, NT-4/5 and ciliary neurotrophic factor have been shown to prevent the loss of facial and spinal motoneurons after axotomy in neonatal rodents.[16–18] Of these, GDNF offers powerful survival-promoting effects on motor neurons *in vitro* and *in vivo*, and promotes nerve regeneration after injury.[19–22] Overexpression of GDNF produces hyperinnervation of neuromuscular junctions,[23] and expression of endogenous GDNF is increased early

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following peripheral nerve injury.[24] The use of GDNF and other neurotrophic factors may prove beneficial for the treatment of peripheral nerve injuries.

Application of neurotrophic factors is hampered by a lack of efficient means of delivery to the peripheral nerves and brain. Direct injection of therapeutic peptides into CNS parenchyma may cause further injury to the CNS. When delivered through oral or systemic routes, these peptides are not feasible due to a short half-life *in vivo* and inability to cross the blood–brain barrier, and often cause harmful adverse effects.[14,15,25] Gene therapy is a strategy for transferring the DNA coding for neurotrophic factors into the CNS or peripheral nerve. This would provide a mechanism for single administration of a vector to achieve limited long term, steady levels of gene product using common routes of delivery.[26,27]

To overcome the problems concerning RLN reinnervation failure, we have demonstrated the potential of gene therapy. Insulin-like growth factor (IGF)-I gene transfer to denervated thyroarytenoid (TA) muscle has been shown to prevent muscle atrophy while improving motor endplate morphology in a rat laryngeal paralysis model,[28,29] and adenoviral vector-mediated GDNF gene transfer prevents motor neuron loss in the nucleus ambiguus in a rat vagal nerve avulsion model.[30] Although our previous studies have demonstrated histological improvements after therapeutic gene transfer, neurofunctional improvement has not previously been clarified. In the present study, to assess the possibility of gene therapy for the recovery of neurological function after RLN injury, we examined the electrophysiological and histological therapeutic effects of adenoviral GDNF gene transfer in an adult rat RLN crush model.

Results

Gene expression in nucleus ambiguus

To investigate whether foreign gene transfer can be achieved by adenoviral vector in the RLN crush model, we first injected adenoviral vector encoding β-galactosidase gene (AxCALacZ) directly into the nerve after left RLN crush injury in adult rats. At 4 days after crush and injection of AxCALacZ, motoneurons of the nucleus ambiguus and associated axons were clearly labeled on X-gal histochemistry (Figure 1). No motoneurons or axons were labeled in the contralateral nucleus ambiguus. We subsequently injected human GDNF gene (AxCAhGDNF) into crushed animals and examined the expression of exogenous GDNF by reverse transcription-polymerase chain reaction (RT-PCR) and GDNF immnohistochemistry. At 4 days after left RLN crush and AxCAhGDNF injection, RT-PCR analysis revealed the expression of virus-induced human GDNF mRNA transcripts in brainstem tissue containing the nucleus ambiguus on the treated side (Figure 2). Virus-induced human GDNF mRNA was not detected on the contralateral side by RT-PCR. At 5 days after left RLN crush and AxCAhGDNF injection, GDNF immnohistochemistry showed expression of virus-induced human GDNF in motoneurons of the nucleus ambiguus on the treated side (Figure 3). No motoneurons or axons were labeled in the contralateral nucleus ambiguus. These results indicate that, after injection to injured axons, virus vectors were delivered to soma of motoneurons through retrograde axonal flow and the virus-encoded foreign gene was successfully induced.

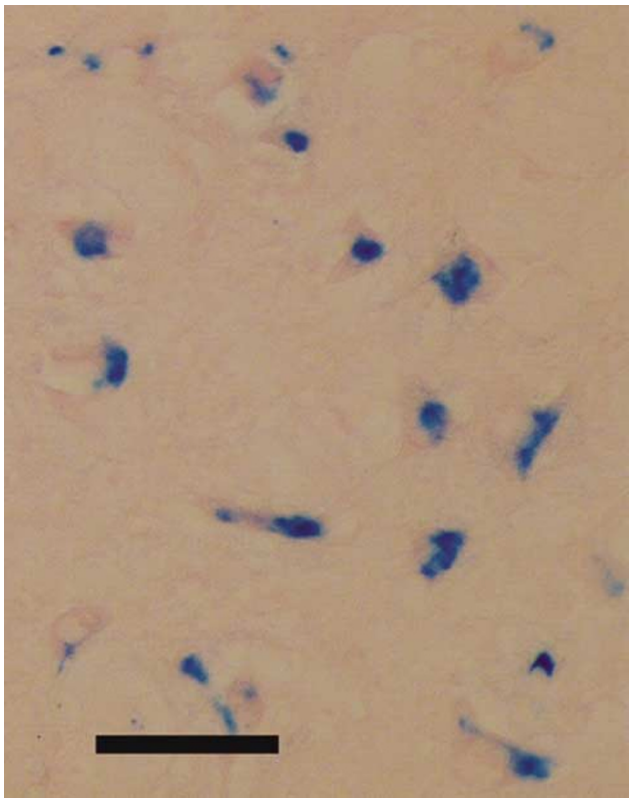


Figure 1 X-gal histochemical staining in the brainstem. At 4 days after RLN crush and intraneural injection of AxCALacZ into the RLN crush site, motor neurons in the nucleus ambiguus were clearly labeled with X-gal. No motoneurons or axons were labeled in the contralateral nucleus ambiguus. Scale bar, 50 μm.

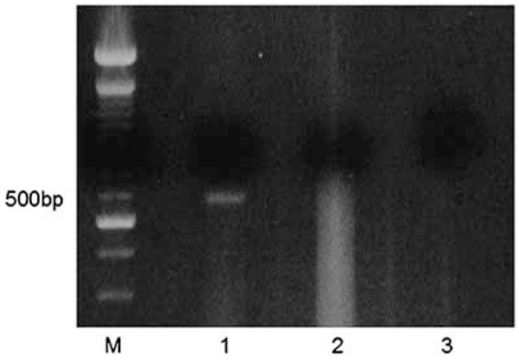


Figure 2 RT-PCR analysis of brainstem tissue including nucleus ambiguus 4 days after RLN crush and AxCAhGDNF treatment. Lane M, size markers; Lane 1, brainstem tissue on the treated side with reverse transcription; Lane 2, brainstem tissue on the treated side without reverse transcription; and Lane 3, brainstem tissue on the contralateral (nontreated) side with reverse transcription. Expression of virus-induced human GDNF mRNA transcripts in brainstem tissue on the treated side. No virus-induced human GDNF mRNA was detected on the contralateral side.

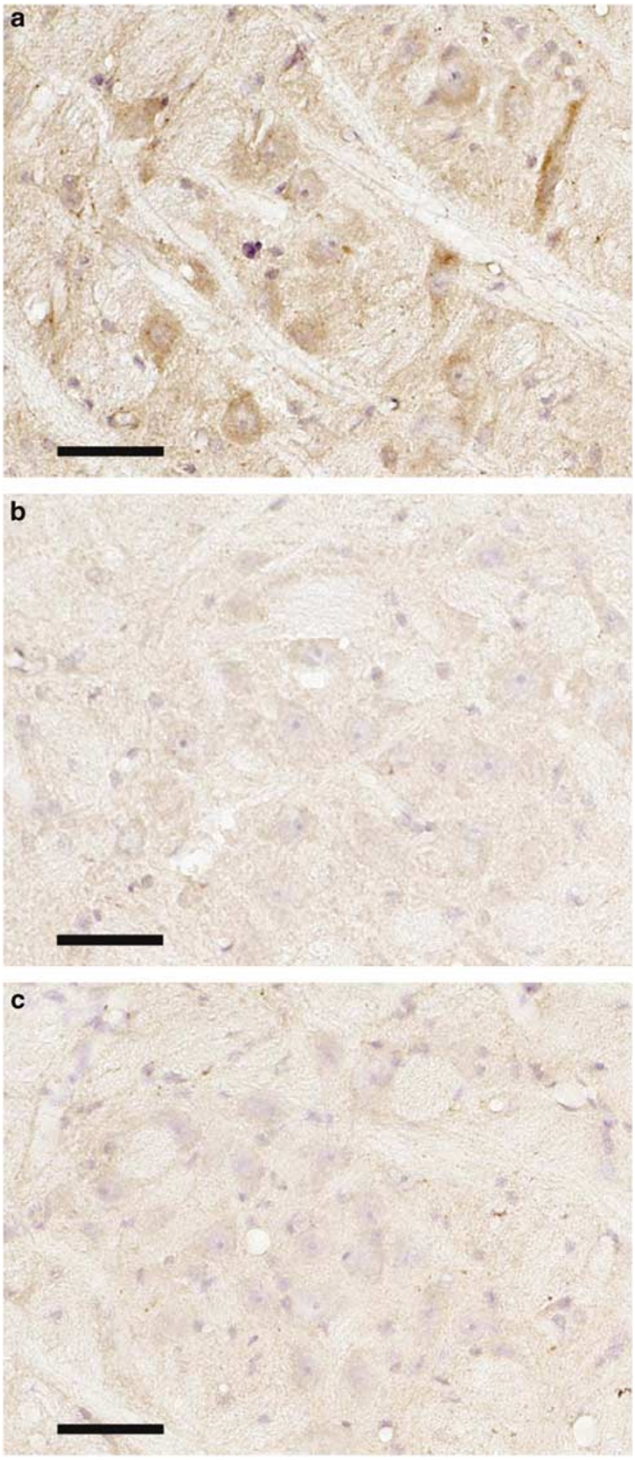


Figure 3 GDNF immunohistochemical staining in the brainstem. (**a**) The nucleus ambiguus on the treated side of AxCAhGDNF-injected animal. (**b**) The nucleus ambiguus on the contralateral (nontreated) side of AxCAhGDNF-injected animal. (**c**) The nucleus ambiguus of RLN crush (control) animal. At 5 days after RLN crush and intraneural injection of AxCAhGDNF into the RLN crush site, motor neurons in the nucleus ambiguus were clearly labeled with GDNF. No motoneurons were labeled in the nucleus ambiguus of contralateral side or RLN crush (control) animal. Scale bars, 50 μm.

Neurofunctional recovery after adenoviral GDNF gene transfer

We injected AxCAhGDNF (Group I), AxCALacZ (Group II), or nothing (Group III) after left RLN crush, and examined the effects of GDNF gene transfer on neurofunctional recovery. Representative electromyography (EMG) findings for Groups I, II and normal RLN are shown in Figure 4. EMG of AxCAhGDNF-injected rats showed shorter latency and shorter time lag of latency compared with AxCALacZ-injected rat, and the potential wave of AxCAhGDNF-injected rats was substantially similar to those in normal rats.

Mean ($\pm$s.d.) motor nerve conduction velocity (MNCV) in Group I was 31.49 ± 7.03 m/s at 2 weeks and 35.59 ± 6.28 m/s at 4 weeks. Normal MNCV of the RLN was 41.38 ± 7.23 m/s ($n = 4$). MNCV of AxCAhGDNF-injected animals thus recovered close to normal levels. MNCV in Group II was 12.59 ± 1.43 m/s at 2 weeks and 19.87 ± 3.05 m/s at 4 weeks, while MNCV in Group III was 17.34 ± 4.25 m/s at 2 weeks and 17.24 ± 3.18 m/s at 4 weeks. At both 2 and 4 weeks after RLN crush, MNCV was markedly reduced in Groups II and III. MNCV was thus significantly faster in Group I than in Groups II and III ($P < 0.05$) (Figure 5), indicating better neurofunctional recovery.

Recovery of vocal fold movement

The number of rats displaying obvious recovery of left vocal fold movement was 4/4 (100%) at 2 weeks and 4/4 (100%) at 4 weeks in Group I, 1/4 (25%) at 2 weeks and 1/4 (25%) at 4 weeks in Group II, and 0/4 (0%) at 2 weeks and 2/4 (50%) at 4 weeks in Group III. Recovery of vocal fold movement was thus significantly better in Group I than in Groups II and III ($P < 0.05$) (Table 1).

Morphometry results of RLN section

Diameters of axons were measured from left RLN sections. Representative cross-sections of RLN samples stained using epon/toluidine blue are shown in Figure 6. Mean diameters of axons and a histogram of axon diameters in normal rats, in Groups I and III are presented in Figure 7. Mean axon diameter was 6.28 ± 0.43 μm in normal rats, 4.76 ± 0.62 μm in Group I and 2.67 ± 0.10 μm in Group III. Mean axon diameter in group I was recovered to almost normal level, significantly larger than in Group III ($P < 0.05$). Histograms of axon diameter demonstrated that Group I displayed a reduction in the number of small axons and an increase in the number of larger axons compared with Group III.

Discussion

Injection of viral vector into nerves has been demonstrated in both spinal cord[31,32] and peripheral nerves.[33,34] The remote injection method of adeno-associated or adenoviral vectors into RLN is reportedly useful for delivering therapeutic gene products to the CNS.[35,36] Virus vector injected directly into the nerve is transported in a retrograde manner to the soma of neurons, and virus-induced foreign genes are thus induced in neurons. Our study showed that by direct injection to the RLN, AxCALacZ could transduce the β-galactosidase gene in the nucleus ambiguus as detected by X-gal histochemistry, and AxCAhGDNF could transduce the

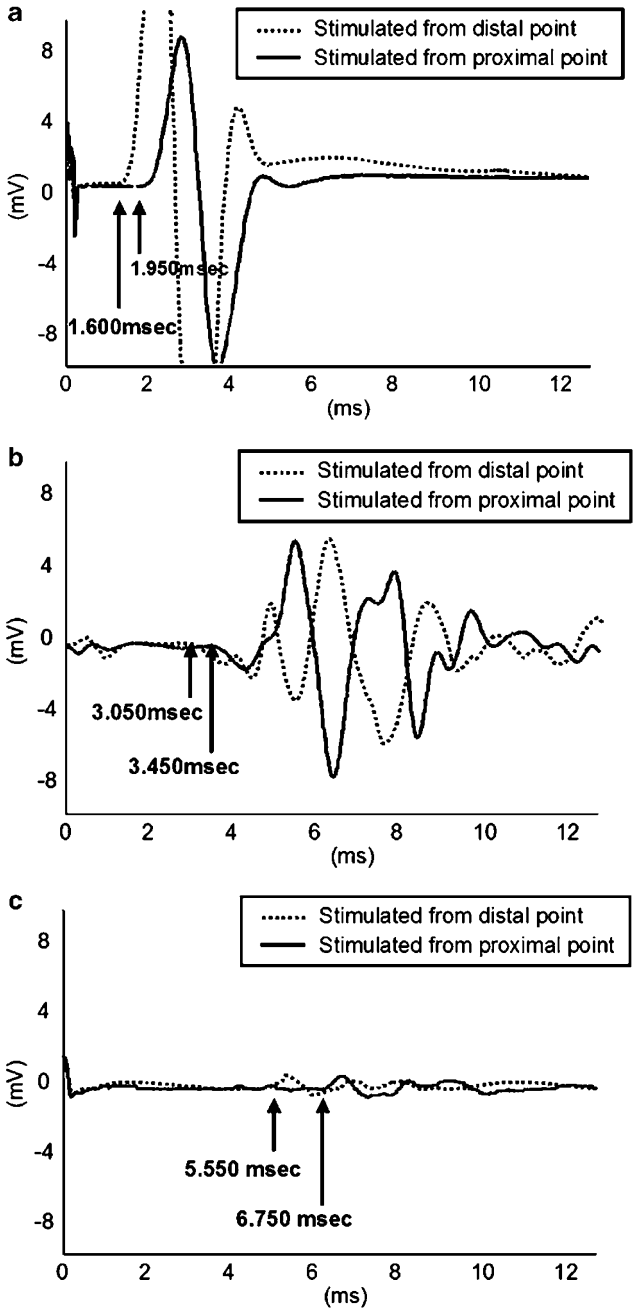


Figure 4 Electromyography of thyroarytenoid muscle. Representative evoked EMG obtained by left RLN stimulation. (**a**) Normal animals, (**b**) AxCAhGDNF-injected animals and (**c**) AxCALacZ-injected animals. EMG in AxCAhGDNF-injected rat revealed big action potential wave and shorter latency (**b**). Conversely, EMG in control rat showed small, unclear action potentials and delayed latency (**c**).

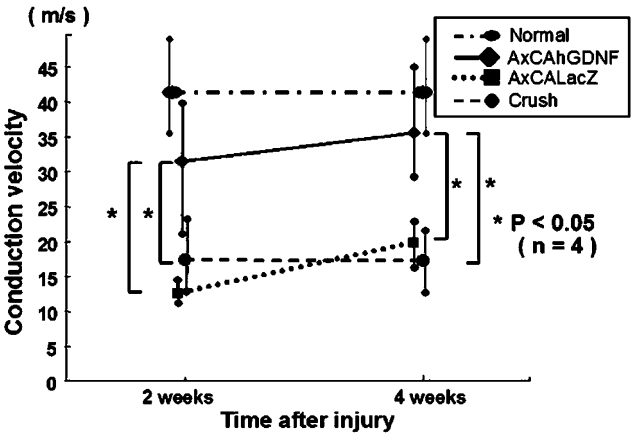


Figure 5 MNCV at 2 and 4 weeks after injury ($n=4$). Significant differences are apparent between AxCAhGDNF- and AxCALacZ-injected animals, and between AxCAhGDNF-injected and RLN crush animals ($P<0.05$) at 2 and 4 weeks after injury. MNCV in AxCAhGDNF-injected animals recovered favorably and was comparable to normal rat. GDNF gene expression offers strong protective and regenerative effects against motor nerve injury.

Table 1 Vocal fold movement recovery in each group

	2 weeks		*4 weeks*	
AxCAhGDNF	4/4	]*	4/4	]*
AxCALacZ	1/4	]	1/4	]
Crush	0/4		2/4	

Numerators show number of animals with vocal fold movement recovery. Denominators show number of animals observed. AxCAhGDNF-injected animals showed significantly better recovery rate for vocal fold movement compared to control animals (AxCALacZ-injected and RLN crush) at 2 and 4 weeks after injury, indicating strong functional recovery following GDNF gene transfer.
*$P<0.05$.

hGDNF gene in the nucleus ambiguus as detected by RT-PCR and GDNF immnohistochemistry.

Remote injection of viral vectors does not appear to cause significant additional neuronal injury. Rubin *et al*.[36] demonstrated that no significant difference existed in percentage of nerve–endplate contacts between an RLN crushed-nerve group and a group with crush injury followed by virus injection. Our data support these findings. MNCV of the RLN did not differ significantly between the crushed-nerve group and the group with crush injury followed by AxCALacZ injection. Although the exact degree of nerve injury caused by the injection process itself is unclear, remote injection of adenoviral vector does not cause significant additional injury after crush and appears to represent a safe way to deliver viral vectors to the CNS.

Nerve crush is a reproducible injury that has been shown to yield a consistent degree of nerve injury.[37,38] This process induces Sunderland second-degree injury (axonotmesis), which involves injury to the axon such that Wallerian degeneration occurs distal to the injury, with only rare and minor features of first- and third-degree injury.[39] This model is ideal for exploring the effects of remotely injected viral vectors, providing an intact nerve for injection and retrograde axonal transport of virus. Recovery is associated with axonal sprouting with the intact individual endoneural tubes of Schwann cell basal lamina.[40]

Bridge *et al*.[38] demonstrated that in the various methods of delivering crush injury in rat sciatic nerve, the functional and histological responses to crush are identical. Their results also showed that MNCV recovered to almost normal levels and only minimal histolo-

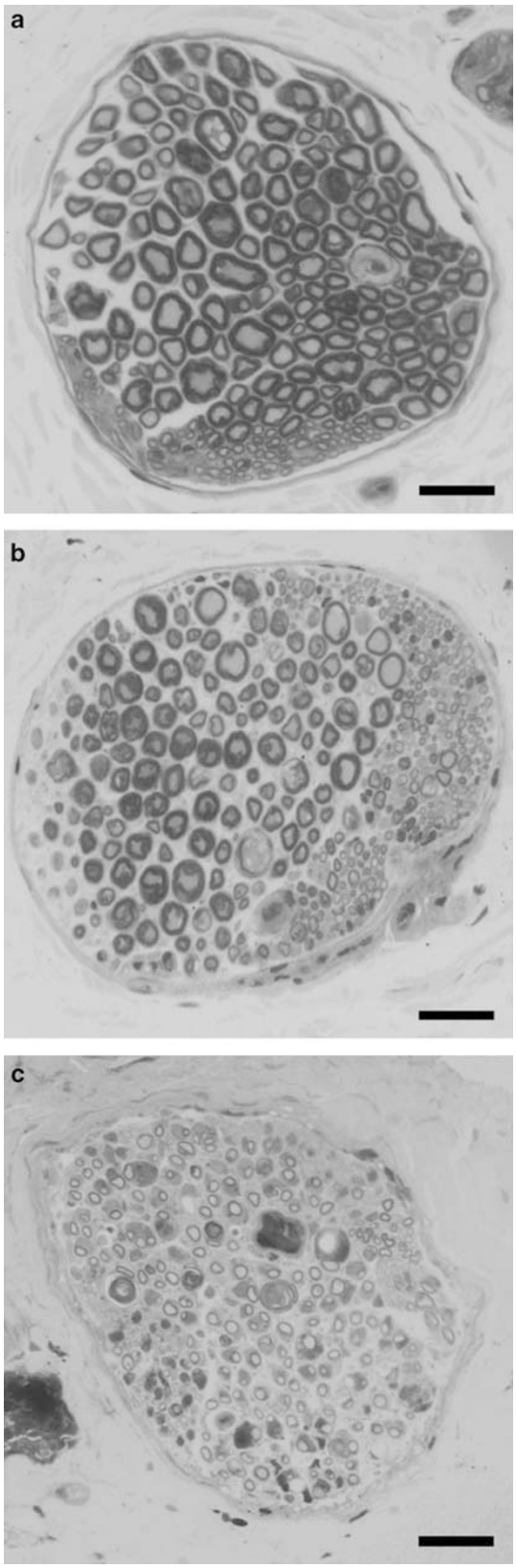


Figure 6 Photomicrographs of sectioned RLN stained with epon/toluidine blue at 5 mm distal from crush site at 2 weeks after injury. (**a**) Normal, (**b**) AxCAhGDNF-injected animal and (**c**) RLN crush animal. Control animal with crush injury displays atrophied and demyelinated axons (**c**). However, in AxCAhGDNF-treated animal (**b**), better myelination and thick axons are observed, comparable to those in normal rat (**a**). Scale bars, 20 μm.

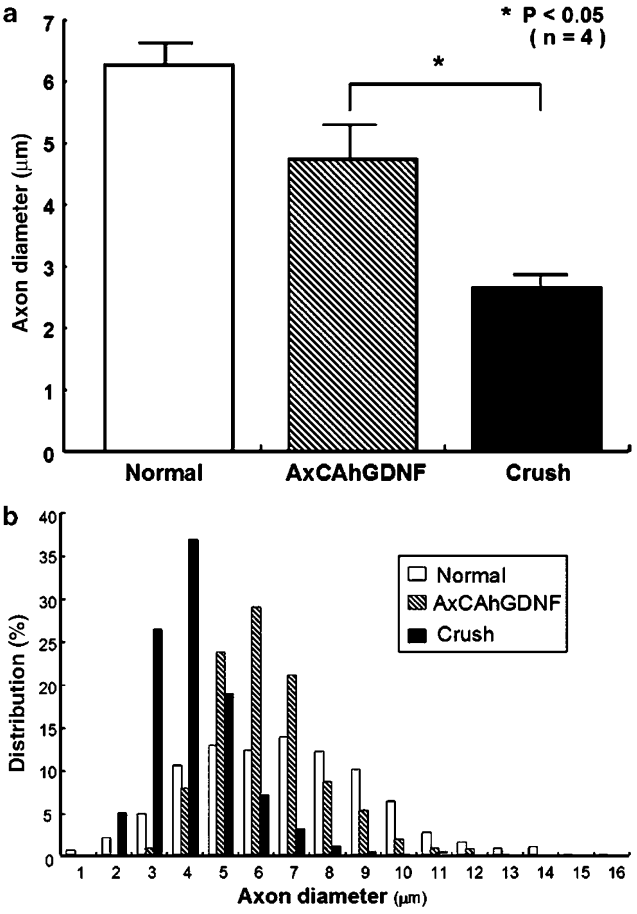


Figure 7 Axonal diameters 2 weeks after injury. (**a**) Mean axonal diameter for each nerve fiber ($n=4$). Significant differences exist between AxCAhGDNF-injected animals and RLN crush animals ($P<0.05$) at 2 weeks after injury, indicating better myelination and axon preservation in AxCAhGDNF-injected animals. (**b**) Distribution of axonal diameters. Histogram shows the total number of axonal diameters in AxCAhGDNF-injected and RLN crush animals ($n=4$). Histogram of diameters reveals a reduced number of small axons and an increased number of larger axons in Group I compared with Group III.

gical abnormalities persisted with every crush method after 8 weeks and that electrophysiological recovery paralleled functional recovery. The present study assessed neurofunctional recovery 2 or 4 weeks after surgery, demonstrating enhancement of motor nerve recovery based on neurofunctional and laryngeal functional data. The timing of assessment was appropriate for the regenerative process of peripheral nerves and our results demonstrate that early neurofunctional recovery might preserve good laryngeal function.

MNCV for the injured nerve is a commonly used physiological measure to evaluate peripheral nerve function in the rat nerve injury model.[37,38,41] In this study, the latency of compound muscle action potential from the TA muscle was recorded, and MNCV was calculated from the latencies of action potential and distance between the two stimulating points. Control animals injected with or without AxCALacZ displayed a markedly slowed MNCV that did not change between 2 and 4 weeks after injury. These results are consistent with limited regeneration of the nerve in control animals that did not improve even up to 4 weeks after injury.

After nerve injury, despite the occurrence of some degree of spontaneous functional recovery, complete functional recovery does not occur due to limited regenerative potential. The primary mechanical trauma causes disruption of the neuron and axon. Damage is due to ischemia at the injury site, shifts in electrolyte concentrations and excessive release of excitatory amino acids.[42] These processes initiate a cytotoxic-signaling cascade and myelin destruction, impeding axonal regrowth. In our study, animals injected with AxCAhGDNF exhibited significant improvements in MNCV compared to control animals, reaching almost normal levels by 2 and 4 weeks after treatment.

We also demonstrated that AxCAhGDNF-injected animals show significant improvements in recovery of vocal fold movement, axonal diameter and myelination, compared to control animals. Thickening of axon diameter and enhancement of myelination are considered to contribute to increases in MNCV. These regenerative effects on the nerve might finally achieve recovery of dynamic laryngeal function (recovery of vocal fold movement).

To minimize the deleterious effects of early trauma and promote and guide axonal regrowth, the delivery of neurotrophic factors has emerged as a promising strategy to manipulate for axonal regrowth in the early phase. This study demonstrated the enhancement of neurofunctional recovery after remote injection of adenovirus vector coding for the GDNF gene into crushed RLN over the course of a few weeks. The vocal folds are extremely delicate structures, and imperceptible injuries can result in excessive vocal complications. Extended injury results in atrophy of the laryngeal muscles, motoneuron loss in the nucleus ambiguus, and decreases in both motor axon density and nerve–endplate contact. Early recovery from axonal degeneration is important for preservation and recovery of laryngeal function. Again, the present methods achieved good preservation and facilitated recovery of laryngeal function.

Laryngeal paralysis most often occurs clinically as a result of vagal nerve or RLN injury after surgical ablation of a tumor involving the head and neck region. If the nerve is injured during surgery, direct injection of the vector into the nerve might prevent paralysis. Alternatively, when paralysis becomes apparent on extubation, the vector can be injected into the nerve after reintubation and opening the wound.

The adenovirus vector was used in this study. For clinical applications, controversy remains regarding the potential risks of virus-mediated gene therapy,[43–45] particularly when applied to nonlethal benign diseases such as laryngeal paralysis. To overcome this problem, safety of the vector must be demonstrated prior to clinical application. Preliminary experiments of nonviral gene transfer systems are also currently underway.

Our previous reports analyzed prevention of motoneuron death[30] and improvements in motor endplate morphology and muscle atrophy using histological methods.[28–30,46] The present study analyzed neurological function using electrophysiological examination *in vivo* and demonstrated neurofunctional recovery. This report carries a higher clinical impact than our previous reports. Adenoviral GDNF gene therapy may save patients with laryngeal paralysis caused by nerve injury during head and neck surgery, by inoculation of the vector at site of nerve damage immediately after injury. These results may open up clinical applications for GDNF gene therapy, not only for laryngeal paralysis, but also for all forms of peripheral motor nerve paralysis.

Materials and methods

Adenovirus preparation

Generation of replication-defective recombinant adenovirus carrying human GDNF cDNA (AxCAhGDNF) and bacterial β-galactosidase gene (AxCALacZ) has been described elsewhere.[46,47] Recombinant adenoviral vectors were propagated and isolated from 293 cells, and purified by two rounds of CsCl centrifugation. Bioactivities of AxCAhGDNF for fetal rat mesencephalic and spinal motoneuron cultures *in vitro* have previously been demonstrated.[46,48]

Animals and surgical procedures

A total of 32 adult male Sprague–Dawley rats (12-weeks old, 340–360 g) were used in these experiments. Animals were anesthetized using ketamine (100 mg/kg, intraperitoneal) and xylazine (10 mg/kg, intraperitoneal) and a midline vertical cervical incision was made through the skin. The submandibular glands and strap muscles were divided to expose trachea, thyroid and larynx. Under a dissecting microscope, the left RLN was exposed just inferior to the left lobe of the thyroid gland. At a point 10 mm proximal to the inferior border of the thyroid gland, the RLN was crushed once with a pair of forceps for exactly 60 s. Crush pressure of forceps was about 80 MPa and crush width was 2 mm. Direct laryngoscopy was performed to confirm left vocal fold paralysis. All procedures were performed in an identical manner by the same investigator using the same equipment.

Rats were randomly assigned to three groups (Groups I–III). In Group I ($n=10$), about 3 μl of AxCAhGDNF (1×10^{10} PFU/ml) was injected at the crush point using a Nanoliter 2000 oocyte microinjector (World Precision Instruments, Sarasota, FL, USA) and micromanipulator immediately after nerve crush. In Group II ($n=10$), AxCALacZ (1×10^{10} PFU/ml) was injected in an identical procedure. In Group III ($n=8$), the RLN was crushed but no viral vector was injected. A glass micropipette was used for direct nerve injection and inserted beneath the perineurium with gentle countertraction of the nerve using microforceps. The strap muscles were repositioned, and the skin was sutured closed. Normal nerve conduction velocity in the RLN was measured using four rats without crush injury or vector injection.

Reverse transcription-polymerase chain reaction

At 4 days after left RLN crush and injection of AxCAhGDNF, animals were euthanized with a lethal dose of ketamine, and brainstem tissue around the nucleus ambiguus was selectively collected under microscopy. Total RNA was isolated from tissue using RNA isolation reagent (Trizol, Invitrogen, Carlsbad, CA, USA) in accordance with the instructions from the manufacturer, and treated with RNase-free DNase (Invitrogen, Carlsbad, CA, USA). First-strand cDNA was synthesized from 380 ng of total RNA using a random primer and Superscript II reverse transcriptase (Invitrogen, Carlsbad, CA, USA) for one PCR analysis.

PCR reactions were performed in PCR buffer containing cDNA template, 200 mM dNTPs, 2 mM MgCl, 20.2 mM of each primer and 25 U/ml of Taq DNA polymerase (TaKaRa, Osaka, Japan). Specific oligonucleotide primers for PCR were designed to amplify adenovirus-derived human GDNF cDNA (forward, 5′-ATGAAGTTATGGGA TGTCGT-3′; reverse, 5′-TCACCAGCCTTCTATTTCTG-3′), which produces 511-bp amplified product.[46] The PCR amplification program comprises 40 cycles of: denaturation at 95°C for 1 min, annealing at 55°C for 90 s and extension at 72°C for 90 s. For negative control reactions, nonreverse transcripted RNA samples and reverse-transcripted brain tissue of nontreated side were processed for PCR. PCR products were subjected to electrophoresis on a 2% agarose gel stained using ethidium bromide.

Histological analysis

For X-gal histochemistry, rats ($n=2$ from Group II) at 4 days after left RLN crush and AxCALacZ injection were anesthetized using ketamine and xylazine, and transcardially perfused with 4% paraformaldehyde in PBS. Brainstem tissue was dissected, immersion-fixed in the same fixative for 2 h, cryoprotected in 30% sucrose in PBS, and serial transverse frozen sections were cut at 16 μm. Cryostat sections were incubated overnight at 37°C in a solution containing 5 mM potassium ferricyanide, 5 mM potassium ferrocyanide, 2 mM magnesium chloride, and 1 mg/ml of 5-bromo-4-chloro-3-indolyl-b-D galactopyranoside (X-gal; Invitrogen) in PBS. Sections were then rinsed and counterstained with eosin.

For GDNF immnohistochemistry, rats ($n=2$ from Group I) at 5 days after left RLN crush and AxCAhGDNF injection were anesthetized and transcardially perfused with 4% paraformaldehyde in PB. Brainstem tissue was dissected, immersion-fixed in the same fixative for 2 h, cryoprotected in 30% sucrose in PBS, and serial transverse frozen sections were cut at 16 μm. Sections were pretreated with 0.3% H_2O_2 in PBS, rinsed in 0.1% Triton X-100 in PBS (T-PBS) and preincubated in 3% normal goat serum in T-PBS. After treatment using an ABC blocking kit (Vector, Burlingame, CA, USA) according to the manufacturer's instructions, sections were incubated overnight at 4°C with rabbit polyclonal antibody to GDNF (Santa Cruz Biotech, Santa Cruz, CA, USA) at a dilution of 1:1000, followed by incubation with biotinylated anti-rabbit IgG at a dilution of 1:100 and ABC reagent (Vector, Burlingame, CA, USA), visualized by 3,3′-diaminobenzidine tetrahydrochloride (DAB)-H_2O_2 solution and counterstained with hematoxylin.

For analysis of RLN section morphometry, left RLNs were excised 2 weeks after surgery and fixed in PBS containing 4% paraformaldehyde/2.5% glutaraldehyde at 4°C for 3 days. Sections were then postfixed in 1% osmium tetroxide, dehydrated through graded ethanol steps, and embedded in Epon 812 (Polysciences, Warrington, PA, USA). At 5 mm distal from the crush site, semithin sections of left RLN at 1 μm thickness were stained using toluidine blue. Digitized images of the entire axonal area were acquired using a CCD camera (PDMC II; Polaroid, Waltham, MA, USA) mounted on an E600 light microscope (Nikon, Tokyo, Japan) with a $\times 100$ objective lens and the entire axonal area of each nerve sampled ($n=4$) was evaluated using Micro Analyzer version 1.1 software (Nihon Poladigital, Tokyo, Japan) on the computer.

Neurofunctional analysis

To assess neurofunctional recovery, rats at 2 or 4 weeks after operation were subjected to measurement of MNCV. Animals were anesthetized and the left RLN was exposed and dissected inferior to the left lobe of the thyroid gland, as described above. The strap muscles were sectioned to expose the larynx, and a laryngeal fissure was made. The left TA muscle was stabbed through the fissure with a needle concentric electrode for recording. To stimulate the left RLN, two bipolar hook electrodes were placed and hooked into the left RLN. One was placed inferior to the left lobe of the thyroid as a distal stimulator, the other at 16 mm proximal to the distal electrode as a proximal stimulator. The nerve was maximally stimulated and compound muscle action potentials in TA muscle were recorded using a Power Lab computer-assisted electromyography machine (AD Instruments, Colorado Springs, CO, USA). Maximal stimulation was achieved by increasing current output until no further change in amplitude of the compound action potential occurred. A current impulse of 0.01-ms duration was delivered. Maximum MNCVs were calculated based on derived latencies and distance between the two stimulating points (16 mm).

At the time of laryngeal fissure creation, recovery of vocal fold movement was also assessed. Recovery was only considered present when equal vocal fold movement on the denervated side was observed compared to the vocal fold on the contralateral nondenervated side. When vocal fold movement on the denervated side was limited (not fixed), recovery was not considered present.

Statistical analysis

Data were analyzed for statistical significance using nonparametric analysis of variance (ANOVA) for comparison among three groups and Mann–Whitney test for comparison between two groups. Values were considered statistically significant at the 5% level.

Acknowledgements

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